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Editors-in-Chief

Autonomous Agents and Multi-Agent Systems

Dear Editors-in-Chief,

Please consider our manuscript, “Same Words, Different Procedure: Counterfactual Transfer Tests for Reusable LLM-Agent Skills,” as a Research article (Regular Paper).

Reusable-skill evaluations commonly report whether an agent succeeds after a skill is supplied. That outcome cannot distinguish benefit from base-agent competence, generic executor assistance, or a misleading surface-similar source. We introduce the Counterfactual Skill Transfer Test (CSTT), a six-condition same-state protocol that crosses source procedural compatibility with lexical similarity and includes no-skill and fair binding-only controls.

Following two retained failed pilots and a final mechanism revision, we froze the method, source assignment, three local model revisions, statistics, decision gates, and a 60-target zero-overlap confirmation roster. Across 180 target-model blocks and 1,080 native episodes, a procedurally matched but lexically far source achieved 97.8% success, versus 10.0% for a lexically near procedural lure. The paired gain was 87.8 percentage points (95% CI 82.8–93.3; task-clustered randomization $p = 1.00 \times 10^{-6}$). Relative to the binding-only control, matched skills produced 150 helpful and no harmful discordances. Three independent audits accepted all episodes and 46,662 native turns without a state-match, native-action, source-assignment, or information-boundary failure.

The manuscript is within the journal’s scope because it provides an evaluation method for agent decision-making architectures, an auditable testbed protocol, and empirical evidence about reliable transfer in autonomous agents. It does not claim to solve retrieval or establish cross-domain transfer; the evidence is limited to the released ALFWorld confirmation and three open model families.

This submission is an independent study and contains no overlapping data, results, figures, tables, or text from another manuscript.

The submitted manuscript is original and is not under consideration elsewhere. It contains no third-party figures or tables requiring permission. The work received no funding, and the authors have no relevant financial or non-financial interests to disclose. Generative-AI assistance is described in the Experimental Design section. The public reproducibility record at doi:10.5281/zenodo.22852634 includes source code, frozen protocols, native interaction logs, audits, tests, and analysis outputs; third-party game data and model weights are not redistributed.

Sincerely,

Liang Song

Corresponding author

Business School, Xi’an International University

Xi’an, Shaanxi 710077, China

liangsong_1976@126.com

On behalf of Liang Song and Zhai Jiabao